

Module 5: The CALCULATE Function

5.1 Concept

CALCULATE is the most powerful DAX function.

It **modifies the filter context** of a calculation, allowing you to create:

- Filtered measures
- Time intelligence calculations
- Conditional aggregations

Syntax:

```
CALCULATE(<expression>, <filter1>, <filter2>, ...)
```

- <expression> : measure or aggregation to calculate
- <filters> : optional conditions or modified context

5.2 Practical Examples

Example 1: Total Sales for a Category

```
Total Fashion Sales :=  
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    Sales[Category] = "Fashion"  
)
```

- ◆ Calculates sum of sales **only for Fashion category**.

Example 2: High Value Sales

```
High Value Sales :=  
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    Sales[SalesAmount] > 5000  
)
```

- ◆ Sums only orders exceeding 5000.

Example 3: Sales Last Year

```
Sales Last Year :=  
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    DATEADD(Sales[OrderDate], -1, YEAR)  
)
```

- ◆ Shifts date context to previous year.

Example 4: Electronics Sales This Month

```
Electronics Sales This Month :=  
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    Sales[Category] = "Electronics",  
    MONTH(Sales[OrderDate]) = MONTH(TODAY())  
)
```

- ◆ Combines multiple filters for dynamic reporting.

5.3 Exercises (Hands-On)

1. Total Electronics Sales

- Measure: Sum SalesAmount where Category = Electronics .

2. High Profit Orders

- Measure: Sum Profit where Profit > 2000 .

3. Fashion Sales Last Year

- Use CALCULATE + DATEADD to sum SalesAmount for Fashion category **last year**.

4. Fast Shipping Electronics

- Sum SalesAmount where:
 - Category = Electronics

- Days to Ship <= 3

5. Non-Discounted Sales

- Sum SalesAmount where Discount = 0 .

5.4 Visualizations in Power BI

1. KPI Cards

- Total Electronics Sales
- High Profit Orders

2. Table/Matrix

- Category | Sales Last Year | Sales This Month

3. Bar Chart

- Products with Fast Shipping (filtered via CALCULATE)

5.5 Exercise Answers (For Trainers)

1. Total Electronics Sales

Total Electronics Sales :=

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    Sales[Category] = "Electronics"  
)
```

- Expect largest sales values in Electronics (Laptops, Phones).

2. High Profit Orders

High Profit Orders :=

```
CALCULATE(  
    SUM(Sales[Profit]),  
    Sales[Profit] > 2000  
)
```

- Only a subset of Electronics and high-value Fashion orders.

3. Fashion Sales Last Year

Fashion Sales Last Year :=

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    Sales[Category] = "Fashion",  
    DATEADD(Sales[OrderDate], -1, YEAR)  
)
```

- Returns total Fashion sales for previous year; smaller than Electronics usually.

4. Fast Shipping Electronics

Fast Electronics Sales :=

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    Sales[Category] = "Electronics",  
    DATEDIFF(Sales[OrderDate], Sales[ShipDate], DAY) <= 3  
)
```

- Only includes orders delivered in 3 days or less.

5. Non-Discounted Sales

Non-Discounted Sales :=

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    Sales[Discount] = 0  
)
```

- Shows total revenue from full-price sales.

✓ By the end of Module 5, students will:

- Understand **CALCULATE** to change filter context.
- Apply multiple filters to measures.
- Combine `CALCULATE` with time intelligence (`DATEADD`) for dynamic reporting.

- Build **real-world metrics** like category-specific, high-value, or filtered sales.